IEEE
AI Team: Task 5

EDA

Summary of Zara's dataset findings and insights

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Introduction

In this report we will explore a ZARA datasheet, the steps taken to check any abnormal values, relationships exploration between different data variables and our humble business insight.

(Any steps mentioned here but dont have screenshots exist in the juptir file.)

Data-set loading and sampling

First we load the data as a CSV file with separation that's ' ;' , then we get the info() of the set to determine the number of features, their data types and how many values are registered as NaN.

We see that for the most part all values are infact not NaN except for very few outliers, so its safe to just remove them since they are of string type so it would be hard to determine what value they actually were.

Then we make sure that we understand what each feature actually mean, like what the ‘origin’ column meaning country of manufacturing and ‘season’ meaning the season for which an item is made.

As last 2 steps of data cleaning we make sure that columns containing strings only contain unique values to represent a given value, ex: only MAN to represent a male and only WOMAN to present a female, and also to check the different types of currencies we have in the dataset.

```
df[df['section'].str.contains(r'^(?!MAN$|WOMAN$)(?:m|M|MALE|male|man|F|f|woman|FEMALE)')]
```

[54] ✓ 0.0s Python

Product ID	Product Position	Promotion	Product Category	Seasonal	Sales Volume	brand	url	name	description	price

```
As we can see there exists no items that arent made for 'WOMEN' or 'MEN' in discription
and no outliers.
```

markdown

```
We will also check the curriencies used to be able to do analysis on prices effictively
```

markdown

```
df['currency'].unique()
```

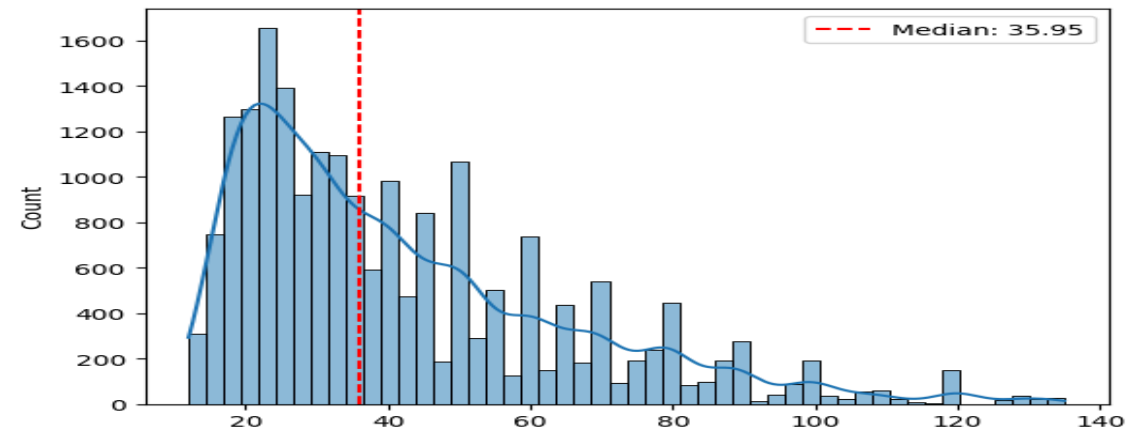
[55] ✓ 0.0s Python

```
<StringArray>
['USD']
```

Univariate Analysis

Exploring the prices of items and the median price of all items

Using a histplot, we will look at the prices of all items of the dataset to look for the median of all prices, what price range customers prefer and what range is too out-landish for most people.

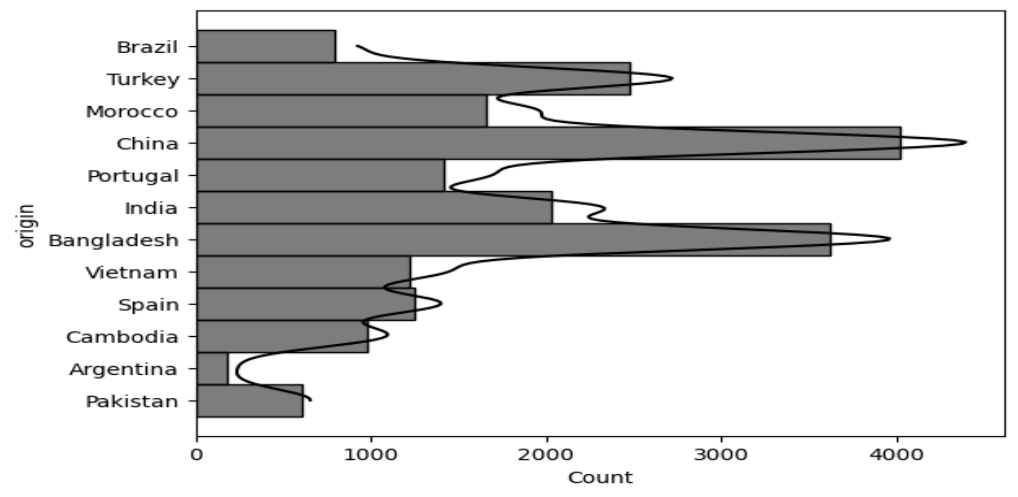


The cost median of all items is about 35.95 dollar, as we can see that the majority of prices fall in the range from 15 dollar up to about 42 dollars, but items 60 and higher and skewing the range with low number of items having such high prices but with a variation of prices big enough to show on the median.

Most productive manufacturing countries

Another univariate to check is which is the single most producing country of zara clothes accoding to the data.

We use a Y- axis hist to visualize this to account for the many number of countries and not having all of them form one big string.

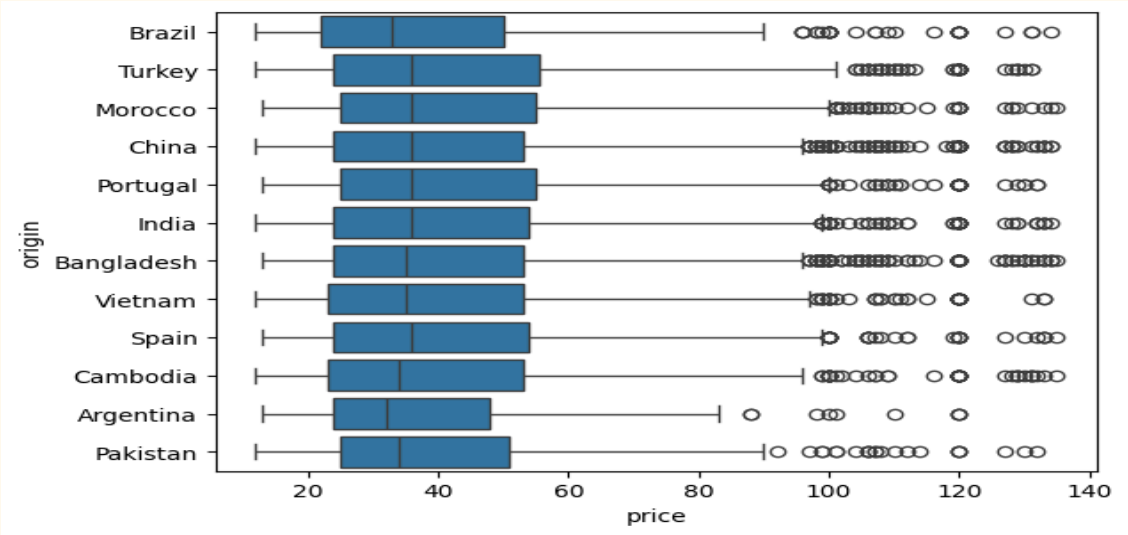


It is apparent that Asian and esp eastern Asian countries are in the lead in terms of manufacturing count, countries like China, Bangladesh and Trukey are the biggest 3, which is logical for their low wage rates.

Bivariate Analysis

The price of clothing made by different countries

We will explore the relationship between the manufacturing countries and prices of clothes using boxplot.



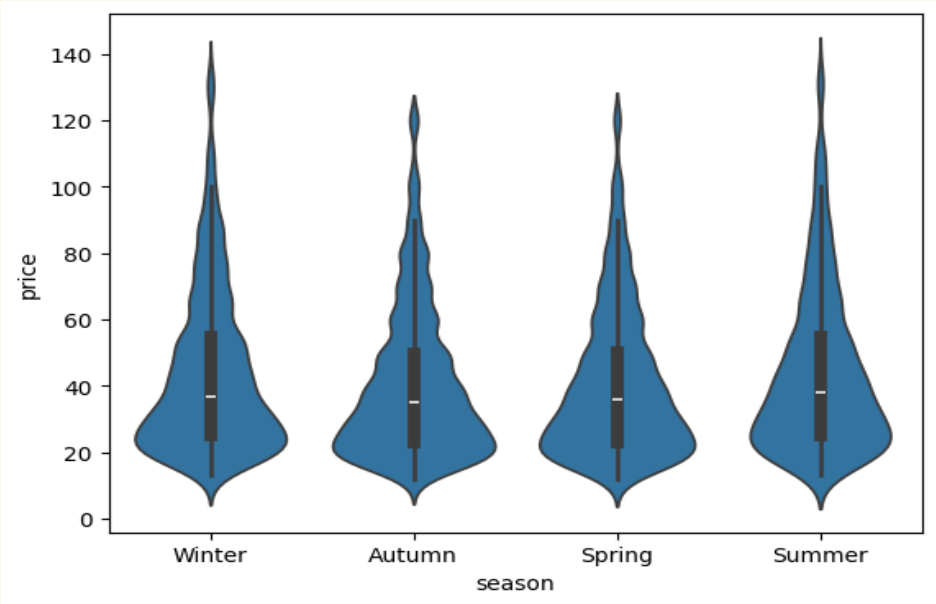
As the graph shows we see that all countries have the same median that is around the actual median of all prices, no country is significantly over-performing relative to the number of pieces it is producing.

Almost every country has a significant number of expensive items reaching nearly 120-140 dollars.

Countries like Paki and India have slightly lower price ranges compared to Turkey or BR.

Seasonal clothes and their prices

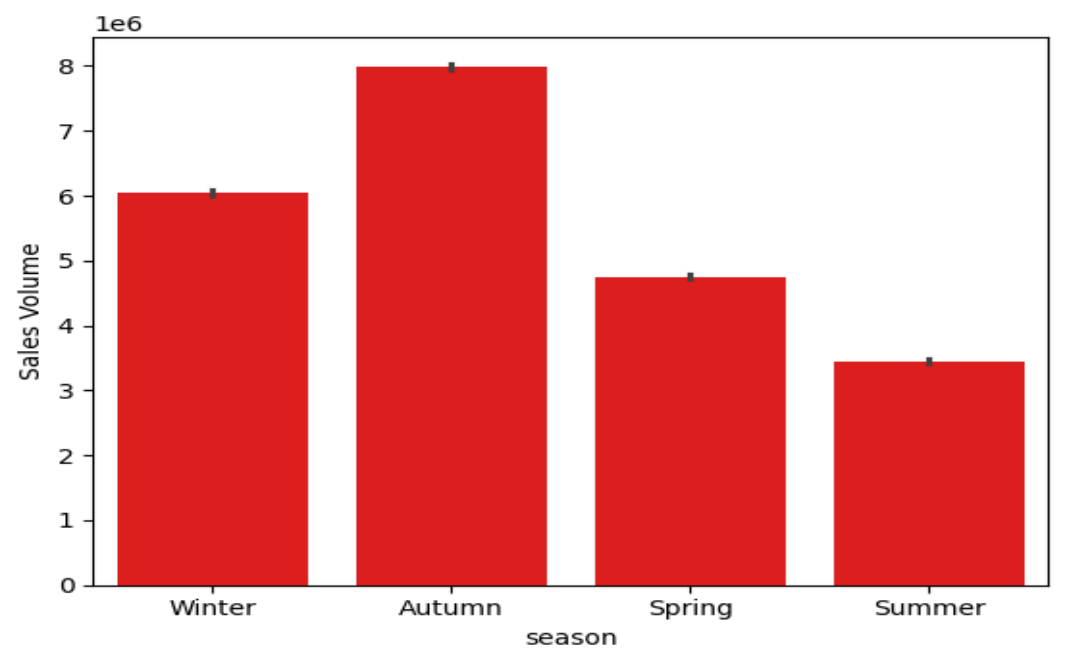
Using a violin plot we will explore how well the brand's clothes perform at each season and how they compare to others using violin-plot.



We see that all seasons share aprox the same mean, while summer and winter most expensive items are slightly more expensive than spring and autmn most expensive ones.

Zara's preferred line

Using a barplot we explore which seasons clothes perform the best.



Surpisingly people perfer ZARA's autmn collection far more than summer's or spring even tho summer is where they usually release their high ticket pieces which tend to be more expensive.

Insights

1- The majority of Zara's sales come from low to mid price range rather than flagship pieces.

2- China and Asia in-general low-pay rate countries are an excellent manufacturing direction for the company.

3- Autumn design are much preferred by the people.

4- Argentina is quite struggling compared to other countries.